

BHUMIBOL DAM, Thailand

WMO: 483770

Lat: 17.2439N

Lon: 99.0025E

Elev: 144

StdP: 99.60

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 14.8	(c) 16.3	(d) 9.7	(e) 7.6	(f) 27.8	(g) 11.3	(h) 8.5	(i) 26.5	(j) 3.4	(k) 28.0	(l) 3.2	(m) 27.7	(n) 0.0	(o) 310	(p) 0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.3	39.3	24.7	38.2	24.7	37.1	24.6	28.4	34.5	27.5	33.4	26.9	32.7	1.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.8	32.5	26.0	21.8	30.7	25.5	21.0	29.6	92.8	34.4	88.4	33.7	85.4	32.6	32.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.8	3.9	3.3	DB	12.5	40.9	1.6	1.3	11.3	41.8	10.3	42.5	9.4	43.2	8.2	44.2
			WB	11.1	29.0	2.0	1.2	9.6	29.9	8.4	30.6	7.3	31.3	5.8	32.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	24.7	27.1	29.9	31.5	30.3	29.1	28.7	28.6	28.3	27.4	26.0	24.2
	DBStd	2.89	2.14	2.20	2.32	2.34	2.52	1.71	1.57	1.40	1.32	1.32	1.78	2.20
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6567	456	478	618	646	629	574	581	577	550	539	480	439
	CDD18.3	3526	198	244	360	396	371	324	322	318	300	281	230	181
	CDH23.3	39488	1649	2683	4857	5577	4699	3853	3687	3554	3142	2530	1880	1378
	CDH26.7	19001	702	1402	2850	3397	2524	1703	1514	1411	1246	977	757	518
Wind	WSAvg	0.5	0.4	0.6	0.7	0.8	0.7	0.8	0.8	0.8	0.4	0.2	0.2	0.2
Precipitation	PrecAvg	1084	8	8	28	63	184	118	96	123	232	183	33	8
	PrecMax	1536	60	89	194	181	444	226	208	236	404	313	150	59
	PrecMin	745	0	0	0	4	34	23	36	48	64	26	0	0
	PrecStd	222	16	18	46	47	116	60	48	46	86	91	36	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	37.6	39.7	41.2	40.3	37.0	35.9	35.6	35.1	34.4	34.4	33.3
		MCWB	21.5	22.1	23.2	25.0	26.0	26.3	26.2	26.1	26.3	26.4	25.4	23.6
	2%	DB	33.3	36.2	38.7	39.8	38.4	35.4	34.4	34.3	34.1	33.3	33.1	32.2
		MCWB	21.8	21.9	23.1	25.2	26.2	26.1	25.9	25.8	26.1	26.1	25.2	23.3
	5%	DB	32.1	35.0	37.6	38.6	36.9	34.1	33.3	33.1	33.1	32.5	32.1	31.1
		MCWB	21.8	22.0	23.1	25.3	26.3	25.9	25.6	25.7	26.0	25.8	24.7	22.8
	10%	DB	30.8	33.7	36.4	37.3	35.3	32.8	32.1	31.9	32.1	31.5	30.9	29.8
		MCWB	21.7	21.9	23.1	25.3	26.1	25.7	25.3	25.3	25.8	25.6	24.3	22.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.9	26.3	28.1	29.5	30.0	28.2	28.1	28.2	28.1	28.0	27.5	26.1
		MCDB	30.6	33.7	35.5	37.3	36.4	33.2	33.0	32.6	32.3	32.3	32.5	31.1
	2%	WB	23.9	24.7	26.3	27.9	28.1	27.1	26.7	26.7	27.0	26.7	26.2	24.8
		MCDB	29.7	31.9	34.2	35.7	34.4	32.3	31.9	31.5	31.9	31.7	31.1	29.9
	5%	WB	23.2	23.9	25.2	26.8	27.0	26.6	26.1	26.2	26.5	26.2	25.6	24.0
		MCDB	29.0	31.3	33.6	34.9	33.7	31.7	31.3	31.1	31.3	31.0	30.3	28.7
	10%	WB	22.5	23.1	24.5	26.0	26.4	26.1	25.6	25.7	26.1	25.8	24.9	23.2
		MCDB	28.3	30.6	32.9	34.4	32.8	31.2	30.6	30.4	30.7	30.4	29.2	27.8
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.7	12.7	11.3	8.8	7.1	6.6	6.9	7.6	8.1	9.8	11.4
		MCDBR	14.0	14.8	14.0	12.4	10.9	8.9	8.3	8.6	9.1	9.6	11.0	12.3
		MCWBR	5.0	4.8	4.2	3.6	3.2	2.9	2.9	2.8	3.1	3.5	4.2	4.6
	5% WB	MCDBR	11.8	13.1	12.4	11.8	9.8	7.8	7.5	7.7	8.2	8.8	9.7	10.4
MCWBR		5.1	5.6	5.0	4.7	3.9	3.1	3.1	3.0	3.3	3.5	4.3	4.6	
Clear-Sky Solar Irradiance	taub		0.467	0.559	0.683	0.642	0.504	0.444	0.433	0.440	0.474	0.544	0.469	0.464
	taud		2.126	1.896	1.637	1.778	2.185	2.382	2.412	2.379	2.269	2.018	2.203	2.157
	Ebn at Noon		814	756	679	711	806	848	858	860	832	758	806	803
	Edh at Noon		149	196	262	228	150	122	119	124	137	171	137	141
All-Sky Solar Radiation	RadAvg		4.69	5.33	5.58	5.99	5.32	4.54	4.22	4.19	4.50	4.58	4.71	4.48
	RadStd		0.22	0.21	0.35	0.34	0.49	0.42	0.39	0.30	0.25	0.27	0.29	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)	+0.26	N/A	N/A	N/A	N/A	+0.11	N/A	N/A	+121	+121

Nomenclature: See separate page